Sandy Bridge Board Shootout

We tested 13 Sandy Bridge motherboards to help you choose the best before you upgrade your system

Biostar TH67XE

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Intel launched its new line of processors – the Sandy Bridge processors – last month. These are the second generation of Core i series of processors that require a completely new motherboard having the LGA 1155 socket, a successor to the old LGA 1156 socket. Also, these new motherboards are based on the new 6-series core-logic chipset. Motherboard manufacturers had already started shipping the Sandy Bridge motherboards, and we thought there wouldn't be a better time to do a motherboard shootout test. We got close to thirteen motherboards from brands such as MSI, ASUS, Gigabyte, Intel, ECS and Biostar. So before we begin the shootout, a quick primer on the features you can expect in this new line of motherboards.

The Two chipsets

As with the previous generation chipsets, the 6-series chipset also comes in two flavours - the higher end P67 chipset and the comparitively economical H67 chipset. The table below encompasses the basic differences between the two chipsets. The major upset, for some, is no scope for using Intel's HD 2000 / 3000 integrated graphics and Quick Sync Video technology on the P67 boards. Another major difference being the fact that P67 board can support upto 3-way SLI or CrossFire setup, whereas H67 board allows use of only one PCIe x16 slot. So the lines are clearly drawn - P67 is for gamers and overclockers whereas H67 boards are ideal for those into audio/video/sound transcoding, editing and for consuming HD media. There is native support for SATA 6 Gbps in the new chipsets, but USB 3.0 is still being supported by third party micro controllers which we found rather strange.

Features	Intel P67	Intel H67
CPU Overclocking via Multiplier	Yes	No
Integrated Graphics Processor output	No	Yes
IGP Overclocking	No	Yes
RAID Support	Yes	Yes
DRAM Speed Option	Upto 2133MHz	Upto 1333MHz

The P67 boards

We got around eight P67 boards from brands like Intel, Gigabyte, MSI, ASUS and ECS. All of them came in the ATX form factor with dark brown colour being common for all the boards. The Gigabyte P67A UD3R comes in a matte dark brown colour which is radically different from the blue Gigabyte motherboards that we are so used to seeing. The heatsinks around the chipset and MOSFETs around the socket region are sturdy gray coloured metal pieces, with a hint of blue, which looks out of place amidst the dark tones.

SATA ports pointing upwards instead of sideways is a big sore on the otherwise clean P67A-UD3R. It supports Dual BIOS, which has a backup BIOS.

The higher end Gigabyte P67A UD7 is an enthusiast Sandy Bridge board. Visually, it appears much sturdier than the P67A-UD3R. The heat sinks around the MOSFETs and P67 chipset are bigger than the ones found on P67A UD3R, and they have a golden colour at the edges. There are four PCI Express x16 slots but their arrangement is such that the graphic card on the first slot, completely covers the second slot. The rear I/O ports section is quite exhaustive on this board.

The MSI P67 boards as well as ASUS P8P67 PRO come with the UEFI BIOS, the graphical UI based BIOS. It allows you to use your mouse and keyboard in the BIOS setup menu. While MSI UEFI comes with icons for each operation like Booting, Overclocking, Utilities, etc, the ASUS UEFI BIOS gives a current status of the board, its temperature and voltage alongwith options to run the processor in Green, Normal mode etc. To go in the advanced mode you will have to click on the top right hand corner tab. Changing options using a mouse is quick and the colour element makes the BIOS screen look quite attractive. Graphical representation of CPU status is great.

ASUS P8P67 PRO is a good board with decent on board features. The angular heatsink looks good and this board has three PCIe x16 slots for hardcore gamers. Absence of dedicated power and reset buttons is not expected from ASUS.

MSI sent three boards namely the P67A-GD65, P67A-GD55 and P67A-C45.

Gigabyte P67A-UD7